

Problem

Traffic police face both planned events (sports matches, festivals, VIP movement, processions, construction activities) and unplanned events (vehicle breakdowns, accidents, water logging, tree falls, protests).

These events often create localized traffic disruption.

The challenge today is that authorities do not have a data-driven way to answer questions such as:

- Which event is likely to create the highest traffic impact?
- Which junctions are most vulnerable during an event?
- How long will the disruption last?
- How many officers and barricades should be deployed?
- Which corridors need proactive monitoring?
- What can be learned from previous similar events?

Most decisions are based on experience rather than historical evidence.

As cities grow and events become more frequent, traffic management requires predictive intelligence rather than reactive response.

Dataset Insight

Our dataset contains exactly the information needed to build an event impact intelligence system:

Event Characteristics

- event_type (planned/unplanned)
- event_cause
- priority
- requires_road_closure

Location Intelligence

- latitude
- longitude
- corridor
- zone
- junction
- police_station

Time Intelligence

- start_datetime
- end_datetime
- created_date
- resolved_datetime

- closed_datetime

Operational Outcomes

- status
- closure information
- resolution duration

This allows us to understand:

- What happened
 - Where it happened
 - How serious it was
 - How long it took to resolve
-

Proposed Solution

Event Impact Forecasting & Response Intelligence System

An AI-powered platform that learns from historical traffic events and helps authorities forecast the operational impact of future events.

Instead of simply recording incidents, the system predicts:

- Expected impact level
 - Expected resolution duration
 - Resource requirements
 - Vulnerable junctions
 - Recommended response actions
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Core Idea

For every new event, the system searches historical events with similar characteristics and generates operational recommendations.

Example:

- Event Type:
- Public Event
-
- Location:
- CBD Area
-
- Priority:
- High
-
- Road Closure:
- No

Output:

- Impact Level:
 - High
 -
 - Expected Resolution:
 - 6.2 Hours
 -
 - Risk Junctions:
 - Queens Statue Circle
 - Cubbon Park Area
 -
 - Recommended Officers:
 - 12
 -
 - Barricades:
 - 18
 -
 - Diversion Planning:
 - Required
-

Module 1: Event Impact Prediction

Objective

Forecast the operational impact of an event before it escalates.

Input

- Event Type
- Event Cause
- Priority
- Corridor
- Zone
- Road Closure Requirement
- Time of Day

Output

- Low Impact
 - Medium Impact
 - High Impact
 - Critical Impact
-

How Impact Is Calculated

Using historical patterns.

For example:

High Impact Indicators

- High priority
- Major corridor
- Public event
- Road closure
- Peak hour

Low Impact Indicators

- Local road
 - Low priority
 - Short-duration breakdown
-

Module 2: Resolution Time Prediction

Objective

Estimate how long traffic disruption will last.

Using:

- $\text{resolution_time} =$
- $\text{resolved_datetime} - \text{start_datetime}$

The model learns:

- Vehicle Breakdown → 40 mins
-
- Water Logging → 3 hrs
-
- Tree Fall → 5 hrs
-
- Public Event → 8 hrs

This helps authorities prepare before congestion grows.

Module 3: Junction Vulnerability Analysis

Using:

- Junction
- Corridor
- Event Frequency

- Resolution Time
- Priority Distribution

The system generates:

Junction Risk Index

Example:

Junction	Risk Score
Hebbal Flyover	9.2
Peenya Junction	8.8
Queens Statue Circle	8.5

Authorities can focus resources on these locations during future events.

Module 4: Hotspot Detection

Using spatial clustering:

- Latitude
- Longitude

The system identifies:

Recurring Event Hotspots

Examples:

- Breakdown Hotspots
- Water Logging Hotspots
- Accident Hotspots
- Event Gathering Hotspots

These become priority monitoring zones.

Module 5: Resource Recommendation Engine

This directly addresses the problem statement.

Based on historical event characteristics, the system recommends:

Manpower

- Low Impact = 2 Officers
-
- Medium Impact = 5 Officers
-
- High Impact = 10 Officers

-
- Critical Impact = 15+ Officers

Barricades

- Road Closure = High Requirement
-
- Public Event = Medium Requirement
-
- Breakdown = Low Requirement

Monitoring Priority

- Immediate
 - Normal
 - Observation Only
-

Module 6: Diversion Planning Assistant

Using:

- Corridor
- Junction
- Zone

the system identifies nearby alternative routes and vulnerable roads.

Instead of waiting for congestion:

- Event Occurs
- ↓
- Affected Junction Identified
- ↓
- Alternative Corridors Suggested

This becomes a decision-support tool for traffic controllers.

Module 7: Post-Event Learning System

One of the official challenges is:

No post-event learning system

Our platform solves this.

For every event:

Store:

- Predicted Impact
- Actual Resolution Time

- Actual Outcome

Measure:

- Prediction Accuracy

Use this information to continuously improve future recommendations.

Innovation

Most systems are:

- Event Reporting Systems

Our system becomes:

- Historical Events
- ↓
- Impact Forecasting
- ↓
- Resolution Prediction
- ↓
- Hotspot Detection
- ↓
- Resource Recommendation
- ↓
- Diversion Assistance
- ↓
- Continuous Learning

Enhancement #1 (Most Important)

Traffic Digital Twin Simulator

This could become your signature innovation.

Instead of:

Event



Prediction

Build:

Event



Virtual City Simulation



Best Response Strategy

Example:

Political Rally

System runs:

Scenario A:

- 5 officers
- No diversion

Congestion Score = 85

Scenario B:

- 10 officers
- Diversion Route A

Congestion Score = 52

Scenario C:

- 12 officers
- Diversion Route B

Congestion Score = 35

Recommended:

Scenario C

This feels futuristic.

Judges remember this.

Enhancement #2

Event Similarity Search Engine

Very few teams will do this.

Use embeddings/vector search.

Input:

Festival Event

System finds:

Top 5 similar historical events

Similarity:

- Location
- Priority
- Time
- Event Type

Output:

"Most similar event occurred on
15-Aug-2024"

Impact:

High

Resolution:

7.1 hrs

Resources Used:

11 Officers

This creates explainable AI.

Judges love explainability.

Enhancement #3

Confidence Score

Current systems:

Prediction:

High Impact

That's it.

Better:

High Impact

Confidence:

91%

Reason:

- Major Corridor
- High Priority
- Peak Hour
- Similar to 42 historical events

Now police can trust the system.

Enhancement #4

Early Warning System

This is extremely valuable.

Before event starts:

System calculates
Risk Index
Example:
Upcoming Festival
Risk Score:
87/100
Trigger:
"Deploy resources 2 hours before start"
Now the system becomes preventive.

Enhancement #5

Counterfactual Intelligence

This is advanced and very impressive.
Instead of:
Impact = High
Show:
What if road closure removed?
Impact:
High → Medium
What if event shifted by 2 hours?
Impact:
High → Low
This is rarely seen in hackathons.

Enhancement #6

Dynamic Resource Optimizer

Current:
High Impact = 10 Officers
Judges may ask:
"Why 10?"
You need optimization.
Use:
Historical Events
to learn:
Minimum resources needed.
Output:
Officers:
9

Barricades:

14

Expected Service Level:

92%

This feels data-driven.

Enhancement #7

Event Chain Prediction

This is genuinely innovative.

Most teams look at one event.

Reality:

One event triggers another.

Example:

Rally

↓

Road Closure

↓

Congestion

↓

Breakdowns

↓

Secondary Accidents

Predict secondary events.

Few teams will think about this.

Enhancement #8

Knowledge Graph

Very powerful for presentations.

Create:

Event

↔ Corridor

↔ Junction

↔ Zone

↔ Resolution Time

↔ Officers

Then ask:

"Show all high-priority public events affecting Hebbal corridor."

Graph AI is uncommon in hackathons.

Instant differentiation.

Enhancement #9

Reinforcement Learning Diversion Planner

Most teams:

Predict route.

Advanced teams:

Optimize route.

State:

Traffic Condition

Action:

Diversion Route

Reward:

Reduced Congestion

This sounds cutting-edge.

Even a simplified prototype impresses judges.

Enhancement #10

LLM Traffic Commander

Instead of dashboards:

Police officer types:

"Tomorrow there is a cricket match in CBD. What should we do?"

System responds:

- Expected Impact: High
- Vulnerable Junctions: X,Y,Z
- Officers Needed: 12
- Barricades Needed: 18
- Diversion Routes Suggested
- Similar Past Events

This creates a "Traffic Copilot".

Judges remember conversational systems.

What I Would Pitch To Win

I would reposition the project as:

Event Impact Intelligence & Traffic Digital Twin

Architecture:

Historical Events



Impact Prediction Model



Resolution Prediction Model



Spatial Hotspot Engine



Event Similarity Engine



Traffic Digital Twin Simulator



Resource Optimization Engine



LLM Traffic Commander



Continuous Learning Loop

Now it looks like a complete AI decision-support platform rather than another ML dashboard.